C1.3 Photosynthesis (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

10 MARKS

1 STATE the name given to the chlorophyll that emits excited electrons in a photosystem. [C1.3.9] [1]

2 STATE two places where photosystems are found. [C1.3.9] [2]

3 STATE the three products of photolysis of water. [C1.3.11] [2]

4 NAME the enzyme that fixes carbon dioxide and the product of the reaction. [C1.3.15] [2]

5 STATE the fraction of triose phosphate converted back to RuBP when glucose is made. [C1.3.17] [1]

6 STATE where protons build up in a chloroplast during the light-dependent reactions. [C1.3.14] [1]

7 STATE where the proton that NADP accepts comes from. [C1.3.13] [1]

Section B: Understanding

14 MARKS

8 EXPLAIN the advantages of arranging many pigment molecules in a photosystem. [C1.3.10] [3]

9 DISTINGUISH between cyclic and non-cyclic photophosphorylation. [C1.3.12] [3]

10 **EXPLAIN** the roles of ATP and reduced NADP in the Calvin cycle. [C1.3.16, C1.3.17] [3]

11 **EXPLAIN** why chloroplasts contain very large amounts of rubisco. [C1.3.15] [2]

12 **OUTLINE** how a plant makes amino acids from what the Calvin cycle produces. [C1.3.18] [2]

13 **STATE** one consequence for the Earth of oxygen released by photosynthesis. [C1.3.11] [1]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

14 [C1.3.15, C1.3.16, C1.3.19]

A suspension of algae was kept in bright light with plenty of carbon dioxide. At time 0 the light was switched off. The concentrations of glycerate 3-phosphate (GP) and RuBP were measured, relative to the values in the light.

TIME / S	−20	0	10	20	40	60
GP	1.0	1.0	1.6	2.0	2.2	2.2
RUBP	1.0	1.0	0.5	0.2	0.1	0.05

a DESCRIBE the change in RuBP after the light was switched off. [2]

b EXPLAIN why GP increased. [2]

c EXPLAIN why RuBP decreased. [2]

d In a second experiment the light stayed on, but carbon dioxide was removed. PREDICT the changes in GP and RuBP, with a reason. [2]

e STATE the two products of the light-dependent reactions that ran out in the dark. [1]

f SUGGEST why the changes began within seconds. [1]

Section D: Longer answer

6 MARKS

15 EXPLAIN how light energy is used to make ATP and reduced NADP in a thylakoid. [C1.3.12–14] [6]